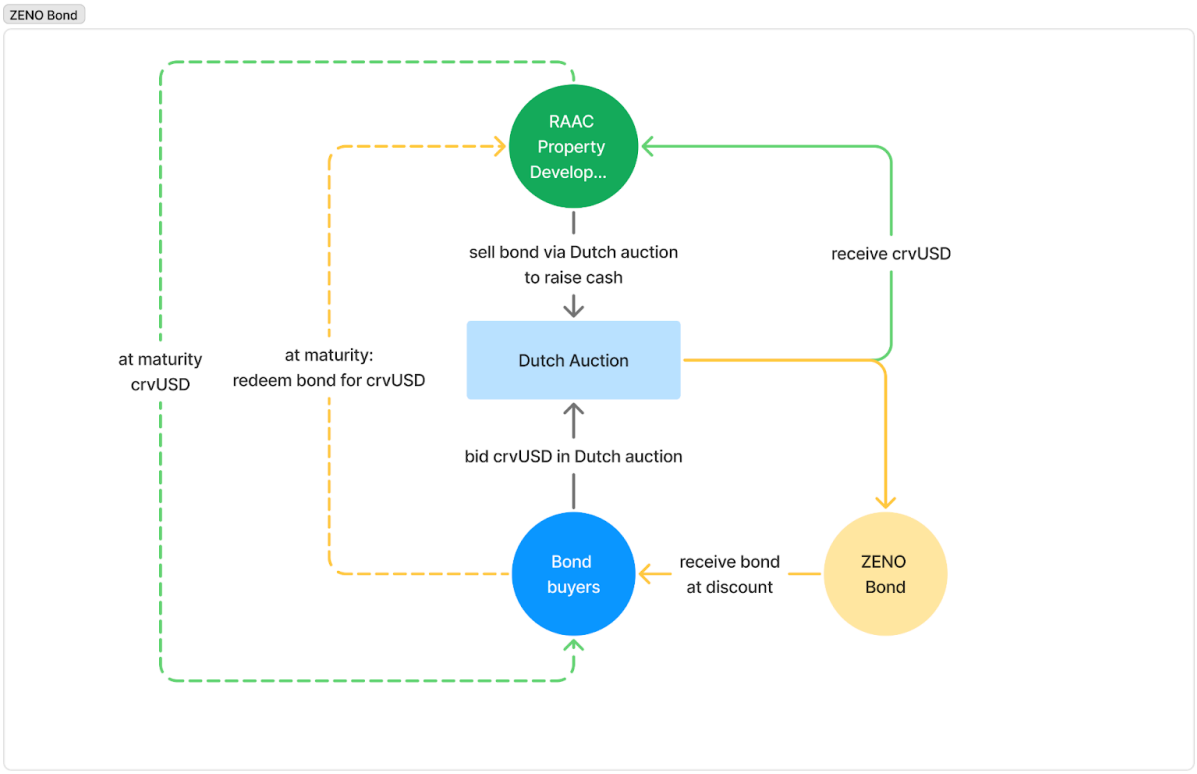
Mechanism Design

ZENO Bonds



[Link]

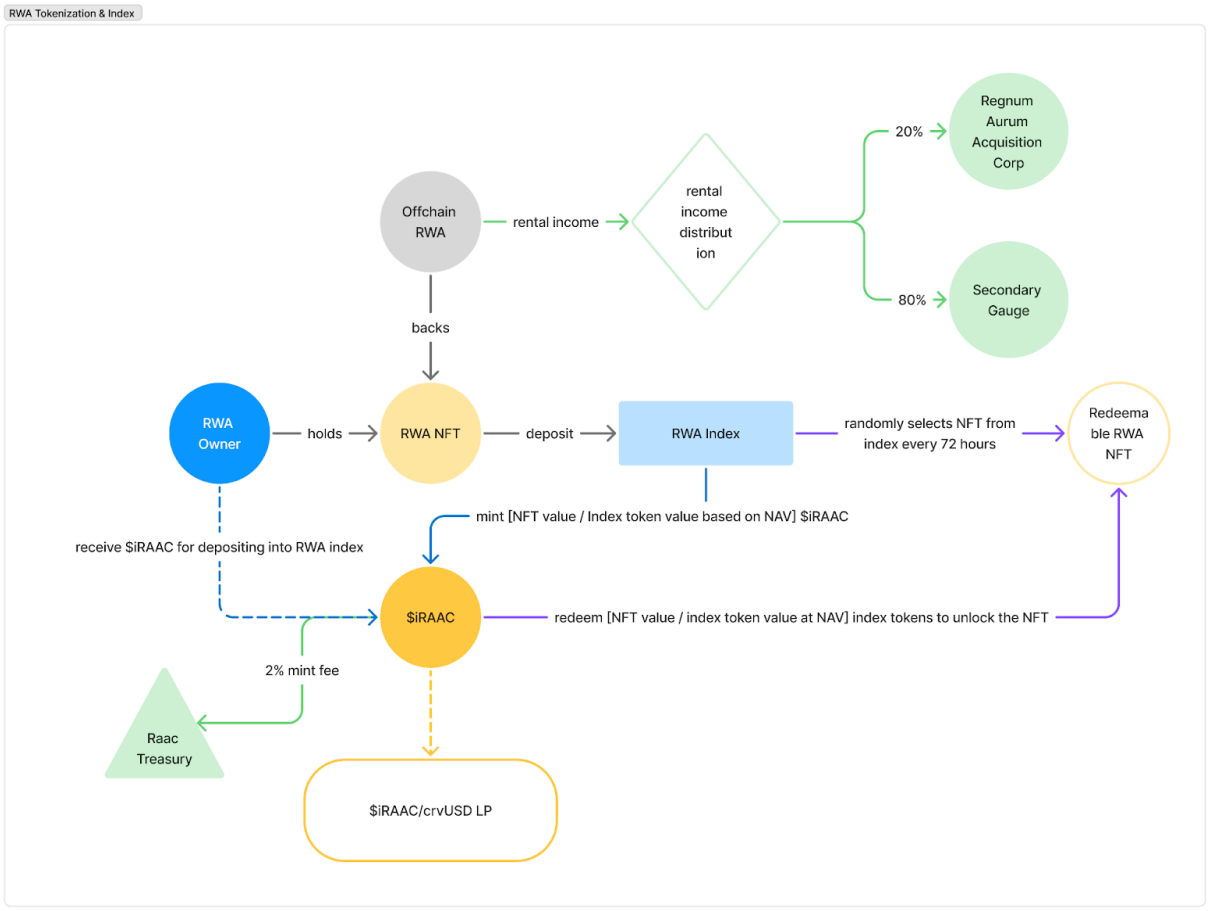
ZENO Bonds (Zero-Coupon External Market Note Operations Bonds) are used discretionally to finance RAAC's operations, such as real estate development, yield strategies, treasury management, or similar.

Buyers purchase ZENO bonds at a discount to their face value at purchase date t_0 and receive the full face value at a defined maturity date $t_{m,}$ where the bond's value is converted into stablecoins. The difference between purchase price and face value represents the buyers' return, which is determined upfront by the discount.

Example

- Purchase price: $70,000
- Face value: $100,000
- Discount: 30%
- Maturity: 5 years
- → Return: $30,000 over 5 years (7.37% per annum)

RWA Tokenization & RWA Index



[Link]

RWA Tokenization

RAAC allows real estate developers to tokenise their offchain property assets as Regna Minima NFTs. This allows developers to borrow against these assets via RAAC's money market in the form of instant, onchain liquidity.

Upon tokenization, the offchain real estate enters RAAC's ownership, with NFT holders able to use their borrowed crvUSD to earn rental yield by placing deposit tokens (LcrvUSD and RcrvUSD) into the liquidity pool as shown in the main value flow diagram.

NFT holders are able to freely trade their asset on third-party marketplaces. NFT holders may also redeem their NFT for the offchain real estate by burning the NFT which unlocks the real estate title. KYC and AML is performed via a zero knowledge proof.

Minting and redeeming will be done by RAAC or whitelisted large-scale real estate developers called Mavericks. The Maverick system decentralizes the offchain to onchain bridge, thereby reducing centralization risk.

RAAC takes care of tokenized real estate through its Repair & Maintenance Fund, with responsibility for property management, maintenance, repair, insurance, payment, and property filings.

80% of the rental income of tokenized real estate goes to RAAC's secondary gauge and will be used to incentivize stability pool depositors in the money market and $iRAAC holders. 20% is directed to the Regnum Aurum Acquisition Corp.

RWA Index

The RWA Index allows retail investors to gain exposure to real estate without having to buy a whole Regna Minima NFT, i.e. an entire tokenised property. The index is created by NFT holders depositing their NFT and receiving index tokens. RAAC may also create and deposit NFTs from their own property holdings. Index tokens represent a share of the real estate (in USD) of the assets held in the index.

NFT depositors receive index tokens ($iRAAC) based on the following formula: *NFT value / Index token value based on NAV.*

RAAC takes a 2% minting fee in the form of index tokens sent to the treasury.

Rental yield associated with index tokens accrues to the RWA gauge directed by $veRAAC holders. The RWA index may receive rental yield from the RWA gauge. Rental yield is then kept in the index, increasing NAV, and is used to buy more real estate. This yield is net of:

- A fee for the RAAC Treasury
- A fee for accruing protocol owned liquidity within RAAC.

Holders may redeem index tokens for an available property, selected from the index (see Redemptions section).

Index tokens can be used as collateral on RAAC to borrow crvUSD. This adds utility and should have a positive impact on index tokens trading closer to NAV.

The Index token price will start at 1$ but may fluctuate up or down, based on NAV.

Example

Real estate worth 1,000,000 USD is added to the index at an index token price of 1$. Therefore, 1,000,000 index tokens get issued. One year later, the real estate is worth 1,100,000 USD. As there are still 1m index tokens in circulation, the price is $1.10

Redemptions
The RWA Index uses a randomized redemption queue where users know which real estate they would get if index tokens are redeemed now. If no redemption happens for 72 hours, the next real estate is randomly selected for redemption.

Holders need to redeem [NFT value / index token value at NAV] index tokens to unlock the NFT.

RAAC Money Market

Interest Rate Framework

RAAC utilizes a soft peg interest rate model which tracks the US Prime Rate. Specifically, below the target rate, borrowers pay half of the US Prime Rate, making borrowing attractive to potential borrowers.

Above the utilization rate target, a penalty interest is introduced which scales linearly with the distance to the target utilization. This provides an incentive for borrowers to repay their loan and allows lenders to withdraw funds, ensuring liquidity. The borrow rate is calculated as follows:

$$R_b = R_b^{target} + max(u \times (U - U_T), 0)$$

with:

- R_b: borrow rate
- R_b^{target}: target borrow rate (½ US Prime Rate)
- U: Utilization
- U_T: Target Utilization
- u: utilization penalty factor

The supply APR (R_s) is calculated from the borrow interest paid by borrowers, minus a reserve factor which constitutes a protocol fee.

$$R_s = R_b \times U \times (1 - RF)$$

with: